

Dow University of Health Sciences

Department of Physiology

BDS Physiology

Learning Objectives and Course Content

(Modules 1–4, 2026)

Course Coordinator: Prof. Dr. Mozaffer Rahim Hingorjo

Program: Bachelor of Dental Surgery (BDS) – First Year

Institute: Dow University of Health Sciences

Academic Year: 2026

Course Information

1	Title of Course	BDS – Human Physiology
2	Course Code	101-PHYB, 102-PHYB, 103-PHYB, 104-PHYB
3	Institute	Dental colleges: DIKIOS, DDC, DIDC (Dow University of Health Sciences)
4	Course Coordinator	Prof. Dr. Mozaffer Rahim Hingorjo
5	Mode of Study	On campus (face-to-face)
6	Duration	1 years
7	Campus / Location Office	DUHS
8	Total Credit Hours	Contact Hours = 300 hours
9	Accredited by	PM&DC, HEC
10	Date of Commencement	12 th January 2026 (Tentative)
11	Date of Evaluation	December 2026 (Tentative; QEC Review)

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Course Details

Introduction

Physiology is the study of biological mechanisms that sustain life. It explains the normal functions of cells, tissues, organs and systems, and provides a foundation for understanding clinical conditions relevant to dentistry.

Teaching and Learning Methods

Lectures: Interactive sessions focusing on core physiological concepts.

Tutorials: Small group discussions to enhance understanding and application.

Practicals: Hands-on laboratory sessions to develop skills and interpretation.

Small Group Discussions: Structured discussions to reinforce concepts and clinical correlation.

Self-Study: Guided independent learning using recommended resources.

Seminars and Symposia: Student-led presentations and academic activities.

Online Learning: Use of LMS for resources, quizzes and assignments.

Problem-Based Learning: Case-based learning using structured problem-solving approaches.

Learning Resources

Guyton and Hall Textbook of Medical Physiology

Human Physiology: From Cells to Systems by Lauralee Sherwood

Physiology by Linda Costanzo

Departmental Physiology Practical Journal

Course Curriculum

MODULE 1: FOUNDATION			
Week	Learning Objectives	Topics	T/L Strategy
Learning Outcome: Describe the principles of homeostasis, body fluid organization, and basic cellular structure and function.			
1	Define physiology and its scope Explain its relevance to dentistry Relate physiological principles to oral health	Introduction to Physiology; scope and relevance to dentistry	Lecture
	Define physiology and its scope Explain its relevance to dentistry Relate physiological principles to oral health	Physiological terminology and levels of organization	Lecture
	Define homeostasis Explain feedback mechanisms Classify control systems	Homeostasis and control systems	Tutorial
	Identify body fluid compartments Describe fluid composition Explain mechanisms of fluid regulation	Body fluid compartments and their regulation	Tutorial
	Identify parts of microscope Describe functions of components Demonstrate proper handling and care	Parts and care of compound microscope	Practical
2	Describe functions of major cell organelles Explain role of organelles in cellular activity Relate organelle function to cell physiology	Functions of Cell and its organelles	Lecture
	Describe structure of cell membrane Explain functions of membrane components Relate membrane structure to its functions	Cell membrane structure and function	Lecture
	Classify types of intercellular junctions Explain basic mechanisms of cell signalling Relate junctions and signalling to cell communication	Intercellular junctions; Cell signalling	Tutorial
	Define osmosis, osmolarity and osmotic pressure Explain movement of water across membranes Relate osmotic concepts to physiological processes	Osmosis, Osmolarity, Osmotic pressure	Tutorial
	Explain principle of osmotic fragility Describe procedure of the test Interpret results in relation to red cell integrity	Osmotic fragility of red blood cells – principle and demonstration	Practical
Learning Outcome: Explain membrane transport mechanisms and basic neurophysiology including nerve conduction and synaptic transmission.			
3	Define simple and facilitated diffusion Differentiate between the two mechanisms Explain factors affecting diffusion	Simple diffusion and facilitated diffusion	Lecture
	Define primary active transport Explain role of ATP in transport Describe function of ion pumps	Active transport: Primary active transport	Lecture
	Define secondary active transport Differentiate symport and antiport Explain role of ion gradients	Active transport: Secondary active transport	Tutorial

	Define filtration and vesicular transport Describe endocytosis and exocytosis Explain role in movement of substances	Bulk transport: Filtration and Vesicular transport	Tutorial
	Compare different transport mechanisms Classify transport processes Apply concepts to physiological scenarios	Revision of membrane transport mechanisms	Tutorial
4	Describe structure of a neuron Classify types of neurons Relate structure to function	Structure and classification of neurons	Lecture
	Define resting membrane potential Explain ionic basis of RMP Describe role of membrane permeability	Resting membrane potential	Lecture
	Explain phases of action potential Describe mechanism of propagation Differentiate types of conduction	Action potential and its propagation	Tutorial
	Describe structure of synapse Explain synaptic transmission Differentiate types of synapses	Synapse: structure and function	Tutorial
	Identify components of PowerLab Describe basic functions Demonstrate proper handling	Introduction to PowerLab	Practical
	Learning Outcome: Explain neuromuscular transmission and skeletal muscle physiology.		
5	Describe structure of neuromuscular junction Explain mechanism of neuromuscular transmission Relate defects to Myasthenia gravis	Neuromuscular junction & Myasthenia gravis	Lecture
	Describe structure of skeletal muscle Explain functional properties of muscle Relate structure to contraction	Structure and properties of skeletal muscle	Lecture
	Explain steps of excitation–contraction coupling Describe role of calcium in contraction Relate electrical events to mechanical response	Excitation–contraction coupling	Tutorial
	Describe sliding filament mechanism Explain role of actin and myosin Relate ATP to muscle contraction	Molecular mechanism of muscle contraction	Tutorial
	Describe principle of muscle twitch recording Demonstrate recording using PowerLab Interpret basic muscle twitch parameters	Recording of simple muscle twitch using PowerLab	Practical
6	Describe sources of energy for muscle contraction Explain concept of oxygen debt Relate energy systems to muscle activity	Energetics of skeletal muscle; Oxygen debt.	Lecture
	Explain length–tension relationship Describe force–velocity relationship Relate mechanics to muscle performance	Mechanics of skeletal muscle contraction	Lecture
	Define motor unit Explain recruitment of motor units Describe muscle tone	Motor unit and muscle tone	Tutorial
	Compare types of muscle Differentiate structural features Relate differences to function	Difference between skeletal, smooth and cardiac muscle	Tutorial
	Describe effects of repeated stimuli Explain muscle fatigue Interpret fatigue patterns in recordings	Effect of repeated stimuli and fatigue in skeletal muscle	Practical

Learning Outcome: Describe the composition and functions of blood, immunity, and mechanisms of hemostasis and their clinical relevance.			
7	Describe components of blood Explain functions of blood Relate composition to function	Functions and composition of blood	Lecture
	Describe stages of erythropoiesis Explain regulation of RBC production Relate role of erythropoietin	Erythropoiesis and regulation	Lecture
	Describe hemoglobin synthesis Explain hemoglobin breakdown Relate iron metabolism to RBC function	Hemoglobin synthesis, degradation and iron metabolism	Tutorial
	Classify types of anemia Explain red cell indices Interpret basic hematological findings	Classification of anemia and red cell indices	Tutorial
	Describe principle of ESR Demonstrate ESR procedure Interpret ESR values	ESR determination	Practical
8	Classify types of anemia Describe features of common anemias Relate causes to physiological changes	Types of anemia	Lecture
	Define polycythemia Differentiate types of polycythemia Explain physiological effects	Polycythemia	Lecture
	Interpret clinical scenarios of anemia Analyze laboratory findings Apply classification to cases	Case studies on anemia	Tutorial
	Classify types of white blood cells Describe functions of WBCs Relate WBCs to immune response	Types of WBCs and their functions	Tutorial
	Describe principle of Sahli's method Demonstrate procedure Interpret hemoglobin values	Hemoglobin estimation – Sahli's method	Practical
9	Define immunity and its types Classify innate and adaptive immunity Describe mechanisms of innate immunity	Immunity and its classification; Innate immunity	Lecture
	Differentiate cell-mediated and humoral immunity Describe roles of lymphocytes Explain antibody-mediated responses	Cell mediated and humoral immunity	Lecture
	Describe principle of DLC Demonstrate differential leukocyte count Interpret DLC findings	To determine differential leukocyte count	Practical
	Describe ABO and Rh systems Explain basis of blood grouping Relate blood groups to compatibility	Blood groups – ABO & Rh System	Tutorial
	Describe principle of cross matching Demonstrate blood grouping procedure Interpret compatibility results	Blood grouping and cross matching	Practical
10	Describe stages of hemostasis Explain platelet plug formation Relate vasoconstriction to hemostasis	Hemostasis – Vasoconstriction & Platelet plug formation	Lecture

	Explain coagulation cascade Describe clot formation Explain clot retraction	Hemostasis – Clot formation and its retraction	Lecture
	Describe fibrinolysis Explain role of plasmin Relate balance between clotting and lysis	Hemostasis – Fibrinolytic mechanisms	Tutorial
	Classify bleeding disorders Explain causes of coagulation defects Interpret basic clinical findings	Disorders of hemostasis	Tutorial
	Describe principles of BT and CT Demonstrate procedures Interpret results	Bleeding time & Clotting time determination	Practical

MODULE 2: CRANIOFACIAL BIOLOGY & DEVELOPMENT			
Week	Learning Objectives	Topics	T/L Strategy
Learning Outcome: Describe the basic organization of neuron and autonomic nervous system and their role in craniofacial and visceral functions.			
1	Define neuron Describe structure of neuron Classify types of neurons	Neuron: overview	Lecture
	Describe organization of ANS Differentiate sympathetic and parasympathetic divisions Identify components of ANS	Autonomic nervous system: organization	Lecture
	Describe neurotransmitters of ANS Explain synaptic transmission in ANS Differentiate adrenergic and cholinergic actions	Autonomic neurotransmission	Tutorial
	Compare effects of sympathetic and parasympathetic systems Explain organ-specific responses Relate ANS actions to physiological functions	Sympathetic and parasympathetic actions	Tutorial
	Relate ANS to salivary secretion Explain ANS role in pupillary and GI functions Apply ANS concepts to clinical scenarios	Applied ANS in craniofacial and visceral functions	Tutorial
	Learning Outcome: Explain the functional organization of cranial nerves and their role in orofacial functions.		
2	Classify cranial nerves Describe functional components Identify major functions	Cranial nerves: overview	Lecture
	Relate cranial nerves to oral functions Explain roles in mastication and salivation Identify nerves involved in swallowing	Cranial nerves in oro-facial functions	Lecture
	Classify functional components Differentiate sensory and motor components Relate components to specific nerves	Cranial nerve components	Tutorial
	Correlate nerve lesions with deficits Interpret clinical findings Apply cranial nerve knowledge	Applied cranial nerves	Tutorial
	Describe steps of examination Demonstrate basic tests Interpret examination findings	Cranial nerve examination	Practical
	Learning Outcome: Explain sensory and motor functions of the trigeminal nerve and their clinical relevance.		

3	Describe sensory functions of trigeminal nerve Identify areas of distribution Relate sensory supply to oral structures	Trigeminal nerve: sensory physiology	Lecture
	Describe mechanisms of pain Classify types of orofacial pain Relate pain pathways to clinical features	Orofacial pain	Lecture
	Describe sensory pathways Differentiate touch and pain pathways Relate pathways to brain regions	Trigeminal pathways	Tutorial
	Demonstrate sensory testing Assess different modalities Interpret examination findings	Sensory examination of trigeminal nerve	Practical
	Correlate lesions with deficits Identify clinical features Apply knowledge to cases	Applied trigeminal dysfunction	Tutorial
4	Describe motor functions of trigeminal nerve Identify muscles of mastication Relate nerve supply to movements	Trigeminal nerve: motor physiology	Lecture
	Describe phases of mastication Explain coordination of jaw movements Relate mastication to oral function	Mastication	Lecture
	Describe jaw reflexes Explain reflex control of mastication Relate reflexes to neural pathways	Masticatory reflexes	Tutorial
	Correlate dysfunction with chewing difficulty Identify clinical features Apply concepts to dental cases	Applied mastication	Tutorial
	Describe functions of facial nerve Identify components of facial nerve Relate nerve to facial functions	Facial nerve: overview	Tutorial
Learning Outcome: Explain the physiological basis of salivation and its neural regulation.			
5	Describe functions of facial nerve Identify components of facial nerve Relate nerve supply to facial and glandular functions	Facial nerve	Lecture
	Describe formation of saliva Explain neural regulation of secretion Relate composition to function	Salivary secretion	Lecture
	Correlate lesions with clinical features Identify effects on facial movements Relate lesions to salivary dysfunction	Facial nerve lesions	Tutorial
	Describe principles of salivary testing Demonstrate assessment of salivary flow Interpret findings	Salivary function testing	Practical
	Relate salivation to oral health Explain effects of altered secretion Apply concepts to clinical scenarios	Applied salivary physiology	Tutorial
Learning Outcome: Explain the physiological mechanisms of swallowing and coordination of oral functions.			
6	Describe mechanism of swallowing Explain coordination of deglutition Relate swallowing to oral function	Swallowing (deglutition)	Lecture
	Identify nerves involved in swallowing Explain roles of cranial nerves Relate nerve function to phases	Cranial nerves in swallowing	Lecture

	Classify phases of swallowing Describe events in each phase Relate phases to control mechanisms	Phases of deglutition	Tutorial
	Correlate dysfunction with dysphagia Identify clinical features Apply concepts to cases	Applied deglutition	Tutorial
	Integrate mastication, salivation and swallowing Explain coordination of oral activities Relate integration to feeding	Integrated oral functions	Tutorial
Learning Outcome: Explain the physiological mechanisms of gastrointestinal secretion, motility, digestion and absorption.			
7	Describe gastrointestinal secretions Explain motility patterns Relate motility to digestion	Gastrointestinal secretion and motility	Lecture
	Describe gastric secretion Explain pancreatic secretion Relate secretions to digestion	Gastric and pancreatic secretion	Lecture
	Describe neural regulation Explain hormonal control Relate regulation to GI activity	Regulation of gastrointestinal functions	Tutorial
	Describe functions of liver Explain bile formation and secretion Relate bile to fat digestion	Liver, bile and gallbladder functions	Tutorial
	Relate GI physiology to clinical conditions Explain basis of common GI disturbances Apply concepts to cases	Applied gastrointestinal physiology	Tutorial
Learning Outcome: Describe the basic physiology of hearing and equilibrium.			
8	Describe digestion of major nutrients Explain role of enzymes Relate digestion to nutrient breakdown	Digestion of carbohydrates, proteins & fats	Lecture
	Describe absorption of nutrients Explain mechanisms of absorption Relate absorption to transport	Absorption of carbohydrates, proteins & fats	Lecture
	Describe mechanism of hearing Explain role of vestibular system Relate functions to balance	Hearing and equilibrium	Tutorial
	Describe principles of hearing tests Demonstrate Weber's and Rinne's tests Interpret test findings	Hearing tests	Practical
	Classify types of hearing loss Explain causes of hearing loss Relate types to clinical features	Hearing loss	Tutorial
Learning Outcome: Explain the basic physiology of vision and its neural control.			
9	Describe image formation in the eye Explain optical components Relate optics to vision	Vision: optics and image formation	Lecture
	Define visual acuity Explain mechanism of refraction Relate refraction to focusing	Visual acuity and refraction	Lecture
	Classify refractive errors Explain causes of errors Relate errors to vision defects	Refractive errors	Tutorial
	Describe principles of visual testing Demonstrate visual acuity assessment	Testing visual acuity	Practical

	Interpret results		
	Describe eye movements Explain pupillary reflexes Relate autonomic control to vision	Eye movements and autonomic reflexes	Tutorial
10	Describe structure of retina Explain phototransduction Relate retinal function to vision	Retina and phototransduction	Lecture
	Describe visual pathways Explain pupillary reflexes Relate pathways to lesions	Visual pathways and pupillary reflexes	Lecture
	Describe principles of color vision testing Demonstrate assessment of color vision Interpret results	Color vision testing	Practical
	Describe principles of field testing Demonstrate visual field assessment Interpret defects	Visual field testing	Practical
	Relate visual physiology to clinical conditions Explain basis of visual defects Apply concepts to cases	Applied visual physiology	Tutorial

MODULE 3: NEUROSCIENCES AND CELLULAR MECHANISMS			
Week	Learning Objectives	Topics	T/L Strategy
Learning Outcome: Explain the organization of sensory systems, including receptors, pathways and perception of sensory modalities.			
1	Classify types of sensory receptors Describe sensory modalities Relate receptors to stimuli	Sensory receptors and modalities	Lecture
	Describe major sensory pathways Differentiate pathways for touch and pain Relate pathways to CNS structures	Somatic sensory pathways	Lecture
	Describe mechanisms of pain Explain types of pain Relate pain to receptors	Pain physiology	Tutorial
	Correlate lesions with sensory deficits Interpret clinical findings Apply pathway knowledge	Sensory pathways: applied aspects	Tutorial
	Demonstrate sensory examination Assess different modalities Interpret findings	Sensory system examination	Practical
Learning Outcome: Explain the role of thalamus and cerebral cortex in sensory processing and interpretation.			
2	Describe functions of thalamus Explain role of somatosensory cortex Relate structures to sensory perception	Thalamus and somatosensory cortex	Lecture
	Define referred pain Explain mechanisms of referred pain Correlate lesions with sensory deficits	Referred pain and sensory lesions	Lecture
	Interpret sensory examination findings Correlate findings with lesions Apply clinical reasoning	Interpretation of sensory findings	Tutorial
	Describe pain modulation pathways	Pain modulation	Tutorial

	Explain role of inhibitory systems Relate modulation to pain perception		
	Apply sensory concepts to clinical scenarios Correlate physiology with symptoms Interpret sensory dysfunction	Applied sensory physiology	Tutorial
Learning Outcome: Explain the organization of motor system, reflexes and clinical correlation of motor lesions.			
3	Describe functions of motor cortex Explain descending motor pathways Relate cortex to voluntary movement	Motor cortex and descending control	Lecture
	Define reflexes Explain reflex integration Relate reflexes to motor control	Reflexes and motor integration	Lecture
	Demonstrate motor examination Assess muscle power and tone Interpret findings	Motor system examination	Practical
	Demonstrate reflex testing Differentiate deep and superficial reflexes Interpret reflex responses	Deep and superficial reflexes	Practical
	Differentiate UMN and LMN lesions Identify clinical features Correlate lesions with signs	UMN and LMN lesions	Tutorial
Learning Outcome: Describe the role of cerebellum and basal ganglia in coordination and control of movement.			
4	Describe functions of cerebellum Explain role in coordination Relate cerebellum to movement control	Cerebellum	Lecture
	Describe functions of basal ganglia Explain role in movement regulation Relate basal ganglia to motor control	Basal ganglia	Lecture
	Demonstrate cerebellar tests Assess coordination and balance Interpret findings	Cerebellar function tests	Practical
	Classify movement disorders Explain physiological basis Relate disorders to neural dysfunction	Movement disorders	Tutorial
	Apply motor concepts to clinical scenarios Correlate physiology with symptoms Interpret motor dysfunction	Applied motor physiology	Tutorial
Learning Outcome: Explain functional organization of cerebral cortex including language and higher cortical functions.			
5	Describe functional areas of cortex Explain roles of major regions Relate areas to functions	Functional areas of cerebral cortex	Lecture
	Describe neural basis of language Explain speech production and comprehension Relate language areas to function	Language functions	Lecture
	Identify cortical areas Relate structure to function Apply localization principles	Cortical localization	Tutorial
	Classify types of aphasia Correlate lesions with deficits Interpret clinical features	Aphasia and cortical lesions	Tutorial
	Apply cortical concepts to cases Correlate physiology with symptoms Interpret cortical dysfunction	Applied cortical physiology	Tutorial

Learning Outcome: Explain the role of hypothalamus in autonomic integration, sleep and regulation of body functions.			
6	Describe functions of hypothalamus Explain autonomic integration Relate hypothalamus to homeostasis	Hypothalamus and autonomic integration	Lecture
	Describe stages of sleep Explain EEG patterns Relate sleep to brain activity	Sleep and EEG	Lecture
	Describe mechanisms of thermoregulation Demonstrate temperature measurement Interpret temperature variations	Body temperature regulation	Practical
	Describe principle of EEG Demonstrate EEG recording Interpret basic patterns	Electroencephalography	Practical
	Classify sleep disorders Explain causes of altered consciousness Relate disorders to physiology	Disorders of sleep and consciousness	Tutorial
Learning Outcome: Explain the principles of endocrine regulation and functions of major endocrine glands.			
7	Describe pituitary hormones Explain hypothalamic control Relate hormones to target glands	Pituitary and hypothalamic hormones	Lecture
	Describe thyroid hormones Explain adrenal hormones Relate hormones to metabolism	Thyroid and adrenal hormones	Lecture
	Explain feedback mechanisms Describe hormonal control Relate regulation to homeostasis	Endocrine regulation	Tutorial
	Relate endocrine function to clinical conditions Explain basis of hormonal disorders Apply concepts to cases	Applied endocrine physiology	Tutorial
	Describe neuroendocrine interactions Explain hypothalamic–pituitary axis Relate integration to body functions	Integrated neuroendocrine control	Tutorial
8	Describe pancreatic hormones Explain regulation of blood glucose Relate insulin and glucagon actions	Endocrine pancreas and glucose homeostasis	Lecture
	Describe roles of PTH, calcitonin and vitamin D Explain regulation of calcium levels Relate hormones to bone metabolism	Parathyroid hormone, calcitonin and vitamin D	Lecture
	Explain regulation of calcium and phosphate Describe hormonal control mechanisms Relate balance to physiological functions	Calcium and phosphate balance	Tutorial
	Classify common endocrine disorders Explain physiological basis Relate disorders to clinical features	Applied endocrine disorders	Tutorial
	Describe role of growth hormone Explain metabolic regulation Relate hormones to growth and development	Growth and metabolic regulation	Tutorial
Learning Outcome: Describe the basic physiology of male and female reproductive systems.			
9	Describe functions of male reproductive system Explain spermatogenesis Relate hormones to male function	Male reproductive physiology	Lecture
	Describe functions of female reproductive system	Female reproductive physiology	Lecture

	Explain oogenesis Relate hormones to female function		
	Describe changes during puberty Explain role of gonadal hormones Relate hormones to sexual development	Puberty and gonadal hormones	Tutorial
	Describe phases of menstrual cycle Explain hormonal regulation Relate physiology to pregnancy	Menstrual cycle and pregnancy physiology	Tutorial
	Relate physiology to clinical conditions Explain basis of reproductive disorders Apply concepts to cases	Applied reproductive physiology	Tutorial
Learning Outcome: Integrate principles of neuroscience, endocrine and behavior in clinical contexts.			
10	Describe components of limbic system Explain role in emotion and behavior Relate limbic system to functions	Limbic system and behavior	Lecture
	Integrate neurophysiological concepts Relate endocrine and neural functions Summarize key mechanisms	Integrated review of neuroscience and endocrine physiology	Lecture
	Analyze clinical scenarios Correlate symptoms with physiology Apply neurophysiological concepts	Case-based neurophysiology review	Tutorial
	Analyze endocrine case scenarios Correlate findings with physiology Apply endocrine concepts	Case-based endocrine review	Tutorial
	Summarize key concepts Integrate module content Apply knowledge to clinical contexts	Comprehensive revision	Tutorial

MODULE 4: STRUCTURE AND FUNCTION OF BODY SYSTEMS			
Week	Learning Objectives	Topics	T/L Strategy
Learning Outcome: Explain the physiological basis of cardiac function, hemodynamics and regulation of blood pressure.			
1	Describe structure of cardiac muscle Explain properties of cardiac muscle Relate conducting system to impulse generation	Cardiac muscle and conducting system	Lecture
	Describe phases of cardiac cycle Explain pressure and volume changes Relate events to heart function	Cardiac cycle	Lecture
	Describe heart sounds Explain mechanism of sound generation Relate sounds to cardiac cycle	Heart sounds	Tutorial
	Demonstrate pulse recording Assess pulse characteristics Interpret findings	Recording arterial pulse	Practical
	Relate cardiac function to clinical conditions Explain basis of common abnormalities Apply concepts to scenarios	Applied cardiac physiology	Tutorial
2	Describe basic principles of ECG Explain components of ECG waves Relate ECG to cardiac activity	Electrocardiography	Lecture

	Define cardiac output and venous return Explain factors affecting cardiac output Relate venous return to cardiac function	Cardiac output and venous return	Lecture
	Demonstrate ECG recording Identify normal ECG components Interpret basic ECG patterns	Electrocardiogram interpretation	Practical
	Explain mechanisms regulating cardiac output Describe role of neural and intrinsic factors Relate regulation to physiological demand	Regulation of cardiac output	Tutorial
	Relate ECG changes to cardiac conditions Explain basis of common abnormalities Apply ECG concepts to cases	Applied ECG physiology	Tutorial
3	Define hemodynamic principles Explain determinants of blood pressure Relate pressure to flow	Hemodynamics and blood pressure	Lecture
	Describe capillary structure Explain mechanisms of exchange Relate Starling forces to fluid movement	Microcirculation and capillary exchange	Lecture
	Demonstrate BP measurement Identify systolic and diastolic values Interpret readings	Recording blood pressure	Practical
	Explain neural mechanisms Describe hormonal regulation Relate regulation to homeostasis	Regulation of blood pressure	Tutorial
	Relate circulation to clinical conditions Explain basis of common disorders Apply concepts to scenarios	Applied circulation	Tutorial
4	Describe regional blood flow Explain factors affecting distribution Relate circulation to organ function	Regional circulation	Lecture
	Describe coronary circulation Explain cerebral blood flow Relate circulation to tissue demand	Coronary and cerebral circulation	Lecture
	Identify types of special circulation Explain unique features Relate differences to function	Special circulations	Tutorial
	Integrate mechanisms of circulation Explain overall cardiovascular control Relate regulation to physiological states	Integrated cardiovascular regulation	Tutorial
	Analyze clinical scenarios Interpret cardiovascular findings Apply physiological concepts	Cardiovascular problem solving	Tutorial
Learning Outcome: Explain the physiological mechanisms of respiration, gas exchange and transport of respiratory gases.			
5	Describe mechanics of breathing Explain role of respiratory muscles Relate pressure changes to ventilation	Mechanics of respiration	Lecture
	Define lung volumes and capacities Describe their physiological significance Relate volumes to lung function	Lung volumes and capacities	Lecture
	Demonstrate spirometry Measure lung volumes Interpret spirometry results	Spirometry	Practical

	Define ventilation and alveolar ventilation Explain factors affecting ventilation Relate ventilation to gas exchange	Ventilation and alveolar ventilation	Tutorial
	Relate mechanics to clinical conditions Explain basis of respiratory abnormalities Apply concepts to scenarios	Applied respiratory mechanics	Tutorial
6	Describe mechanism of gas exchange Explain diffusion across alveolar membrane Relate exchange to partial pressure gradients	Gas exchange in lungs	Lecture
	Describe transport of oxygen Explain transport of carbon dioxide Relate transport to hemoglobin function	Transport of oxygen and carbon dioxide	Lecture
	Identify factors influencing gas exchange Explain effects of ventilation–perfusion mismatch Relate factors to oxygenation	Factors affecting gas exchange	Tutorial
	Describe neural control of respiration Explain chemical regulation Relate regulation to respiratory drive	Regulation of respiration	Tutorial
	Relate gas transport to clinical conditions Explain basis of hypoxia Apply concepts to scenarios	Applied gas transport	Tutorial
7	Define hypoxia and cyanosis Classify types of hypoxia Relate causes to clinical features	Hypoxia and cyanosis	Lecture
	Describe respiratory adaptations Explain responses to exercise and altitude Relate adjustments to physiology	Respiratory adjustments in altered states	Lecture
	Interpret respiratory parameters Correlate findings with physiology Apply clinical reasoning	Interpretation of respiratory findings	Tutorial
	Relate respiratory physiology to clinical conditions Explain basis of respiratory disorders Apply concepts to scenarios	Applied respiratory physiology	Tutorial
	Integrate respiratory concepts Summarize key mechanisms Apply knowledge to cases	Integrated respiratory review	Tutorial
Learning Outcome: Explain the mechanisms of renal function, body fluid regulation and acid–base balance.			
8	Describe renal blood flow Explain mechanism of glomerular filtration Relate factors affecting GFR	Renal circulation and glomerular filtration	Lecture
	Describe tubular reabsorption Explain mechanisms of secretion Relate processes to urine formation	Tubular reabsorption and secretion	Lecture
	Describe principles of urine analysis Demonstrate urine examination Interpret findings	Urine examination	Practical
	Explain handling of key solutes Describe transport mechanisms Relate handling to kidney function	Renal handling of substances	Tutorial
	Relate renal function to clinical conditions Explain basis of renal disorders Apply concepts to scenarios	Applied renal physiology	Tutorial

9	Describe mechanisms of urine concentration Explain dilution of urine Relate processes to ADH	Concentration and dilution of urine	Lecture
	Explain regulation of osmolarity Describe control of ECF volume Relate hormones to fluid balance	Regulation of extracellular fluid osmolarity and volume	Lecture
	Describe counter-current system Explain role in urine concentration Relate mechanism to medullary gradient	Counter-current mechanism	Tutorial
	Explain regulation of water balance Describe causes of edema Relate fluid imbalance to clinical states	Water balance and edema	Tutorial
	Integrate renal mechanisms Explain overall fluid regulation Apply concepts to clinical scenarios	Integrated renal regulation	Tutorial
10	Describe mechanisms of acid–base regulation Explain role of buffers, lungs and kidneys Relate pH regulation to homeostasis	Acid-base balance	Lecture
	Describe mechanism of micturition Explain neural control of bladder Interpret cystometrogram	Micturition and cystometrogram	Lecture
	Classify acid–base disorders Interpret arterial blood gas changes Relate disorders to physiological mechanisms	Acid-base disorders	Tutorial
	Relate renal physiology to clinical conditions Explain basis of fluid imbalance Apply concepts to scenarios	Applied renal physiology and body fluid balance	Tutorial
	Integrate concepts across systems Summarize key physiological mechanisms Apply knowledge to clinical cases	Integrated review - cardiovascular, respiratory & renal physiology	Tutorial

Assessment Methodology

For a modular design, it's best to align **assessment methods with the learning environment** (not just content). Each teaching format tests a *different level of Bloom's taxonomy*, so the assessment should reflect that.

1. Lecture → Assessment Methods

(Focus: understanding, integration, application)

Recommended methods:

- **MCQs (single best answer)**
 - Best for conceptual clarity and discrimination
 - Mechanism-based MCQs → ideal here
 - **Short Answer Questions (SAQs)**
 - Test explanation of mechanisms
 - e.g., *Explain how GFR is autoregulated*
 - **Structured viva (targeted)**
 - Short, focused questioning rather than open viva
 - **Concept mapping / flowcharts**
 - Especially useful for physiology pathways
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2. Tutorial → Assessment Methods

(Focus: application, reasoning, discussion)

Recommended methods:

- **Case-based MCQs**
 - Short scenario + lab values
- **Structured case discussion**
 - Students explain mechanisms stepwise
 - Can be graded with a checklist
- **Think-pair-share / group tasks**
 - Assess participation + reasoning
- **Problem-solving exercises**
 - e.g., interpret ABG, predict physiological changes
- **Mini quizzes (formative)**

- o 3–5 MCQs at end of session

Key feature:

- ✓ Assessment should be **interactive + formative**, not just summative
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3. Practical → Assessment Methods

(Focus: skills, interpretation, demonstration)

Recommended methods:

A. OSPE (Objective Structured Practical Examination) — GOLD STANDARD

- Stations-based:
 - o Instrument identification
 - o Graph interpretation
 - o Data analysis
 - o Short procedure steps

- ✓ Highly structured → minimizes examiner bias
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B. Direct observation (checklist-based)

- Performance of:
 - o BP measurement
 - o Spirometry
 - o Hematology procedures
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C. Practical viva (structured)

- Linked to experiment performed
 - Avoidance of random questioning
-

D. Record book / logbook

- Completion + accuracy + interpretation
-

E. Data interpretation questions

- Graphs, tracings, lab values
- Very important for physiology